ASE CHAPTER 2

T1 MCQ Question Pack · Life-Cycle Models

25 Companion Slide MCQs with Content-Derived Solutions and Analytical Rationales

This pack contains the **25 multiple choice questions** from K.K. Aggarwal & Yogesh Singh (*Software Engineering*, 3rd ed., pp. 37–39, and companion slides 41–45). Note on Official Key: The official textbook answer key on printed p. 499 confirms the keys (2.1 to 2.25), cross-referenced with chapter prose and selection matrices.

Question 2.1

Spiral Model was developed by: - (a) Bev Littlewood - (b) Barry Boehm - (c) Roger Pressman - (d) Victor Basili

Correct Answer: (b) Barry Boehm

Technical Rationale: The spiral model was formulated and published by Barry Boehm in 1986 to incorporate risk analysis and management into the software development process.

> **ملاحظة الطالب والفخ الامتحاني:** - باري بوم (Barry Boehm) هو صاحب فكرة النموذج الحلزوني ونموذج تقدير التكاليف الشهير COCOMO. احذر الخلط مع روجر بريسمان (مؤلف الكتب) أو فيكتور باسيلي. (تنبيه: ورد اسمه مطبوعاً في الكتاب الأصلي ص. 37 والشريحة 41 بالخطأ المطبعي "Berry Boehm"، والمفتاح ص. 499 يعتمد الخيار (b)، والصحيح علمياً هو Barry Boehm).

Source: K.K. Aggarwal & Singh (2007), §2.4, p. 27; Printed Question p. 37 (Q2.1) & Slide 41; Official Answer Key p. 499.

Question 2.2

Which model is most popular for student's small projects? - (a) Waterfall model - (b) Spiral model - (c) Quick and fix model - (d) Prototyping model

Correct Answer: (c) Quick and fix model

Technical Rationale: The quick and fix model (also referred to as Build & Fix) is the most popular approach for small student programming exercises of 100 to 200 lines because code is written immediately without requiring formal specifications or architectural design.

> **ملاحظة الطالب والفخ الامتحاني:** - يُطلق عليه في مراجع أخرى Build and Fix؛ هو أسلوب عفوي (Ad-hoc) يناسب التمارين الفردية الصغيرة ولا يصلح للمشاريع الحقيقية.

Source: K.K. Aggarwal & Singh (2007) Slides 4–5; Slide 41 (Q2.2).

Question 2.3

Which is not a software life cycle model? - (a) Waterfall model - (b) Spiral model - (c) Prototyping model - (d) Capability maturity model

Correct Answer: (d) Capability maturity model

Technical Rationale: Waterfall, Spiral, and Prototyping are software development process/life-cycle models defining phases of development. The Capability Maturity Model (CMM) is an organizational process improvement and maturity appraisal framework developed by the SEI, not an SDLC process model.

> **ملاحظة الطالب والفخ الامتحاني:** - نموذج CMM يقيّم نضج المنظمة البرمجية ككل (Maturity Framework)، ولا يحدد مراحل بناء البرمجيات كدورة حياة.

Source: K.K. Aggarwal & Singh (2007) Slide 41 (Q2.3).

Question 2.4

Project risk factor is considered in: - (a) Waterfall model - (b) Prototyping model - (c) Spiral model - (d) Iterative enhancement model

Correct Answer: (c) Spiral model

Technical Rationale: The primary defining feature of the spiral model is explicit, structured risk management. Quadrant 2 of each 360-degree loop is specifically dedicated to identifying, analyzing, and resolving technical and management risks.

> **ملاحظة الطالب والفخ الامتحاني:** - كلما اقترن السؤال بعامل المخاطرة (Project Risk / Risk Analysis)، فالإجابة القطعية هي Spiral Model.

Source: K.K. Aggarwal & Singh (2007) Slide 19, 21; Slide 41 (Q2.4).

Question 2.5

SDLC stands for: - (a) Software design life cycle - (b) Software development life cycle - (c) System development life cycle - (d) System design life cycle

Correct Answer: (b) Software development life cycle

Technical Rationale: Within software engineering, SDLC standardly denotes the Software Development Life Cycle, representing the structured phases from software conception to delivery and maintenance.

> **ملاحظة الطالب والفخ الامتحاني:** - في سياق هندسة البرمجيات، الحرف S يرمز لـSoftware و D لـDevelopment.

Source: K.K. Aggarwal & Singh (2007) Slide 3; Slide 41 (Q2.5).

Question 2.6

Build and fix model has: - (a) 3 phases - (b) 1 phase - (c) 2 phases - (d) 4 phases

Correct Answer: (c) 2 phases

Technical Rationale: The Build & Fix model is defined as a simple two-phase ad-hoc model: write code (Build), and then modify code to repair errors (Fix).

> **ملاحظة الطالب والفخ الامتحاني:** - نص السلايد 4 واضح تماماً: "Simple two phase model: Build Code ⟵ Fix".

Source: K.K. Aggarwal & Singh (2007) Slide 4; Slide 42 (Q2.6).

Question 2.7

SRS stands for: - (a) Software requirements specification - (b) Software requirements solution - (c) System requirements specification - (d) None of the above

Correct Answer: (a) Software requirements specification

Technical Rationale: SRS standardly stands for Software Requirements Specification, the formal document produced at the conclusion of requirements analysis detailing what a system will do.

> **ملاحظة الطالب والفخ الامتحاني:** - الـSRS هي الوثيقة التعاقدية الرسمية التي تصف "ماذا" يفعل النظام، وتثبت المتطلبات الوظيفية وغير الوظيفية.

Source: B.B. Agarwal et al. (2010), §3.2.2, p. 57; K.K. Aggarwal & Singh (2007) Slide 10, 42 (Q2.7).

Question 2.8

Waterfall model is not suitable for: - (a) Small projects - (b) Accommodating change - (c) Complex projects - (d) None of the above

Correct Answer: (b) Accommodating change

Technical Rationale: A major recognized weakness of the classical waterfall model is that requirements are frozen early and development proceeds strictly linearly, making it unsuitable for accommodating frequent changes during project execution.

> **ملاحظة الطالب والفخ الامتحاني:** - السلايد 8 ينص حرفياً: "This model is not suitable for accommodating any change". كما أنه ضعيف في المشاريع الكبيرة، لكن عدم استيعاب التغيير هو العيب المنهجي الأبرز.

Source: K.K. Aggarwal & Singh (2007) Slide 8; Slide 42 (Q2.8).

Question 2.9

RAD stands for: - (a) Rapid application development - (b) Relative application development - (c) Ready application development - (d) Repeated application development

Correct Answer: (a) Rapid application development

Technical Rationale: RAD denotes Rapid Application Development, a model designed to reduce development time through rapid prototyping, active user participation, and component reuse.

ملاحظة الطالب والفخ الامتحاني: - اختصار RAD يعني Rapid Application Development (تطوير التطبيقات السريع).

Source: K.K. Aggarwal & Singh (2007) Slide 12; Slide 42 (Q2.9).

Question 2.10

RAD model was proposed by: - (a) Lucent Technologies - (b) Motorola - (c) IBM - (d) Microsoft

Correct Answer: (c) IBM

Technical Rationale: The Rapid Application Development model was developed by IBM in 1980.

ملاحظة الطالب والفخ الامتحاني: - تنص شريحة المصدر 12 بوضوح: "Developed by IBM in 1980".

Source: K.K. Aggarwal & Singh (2007) Slide 12; Slide 42 (Q2.10).

Question 2.11

If requirements are easily understandable and defined, which model is best suited? - (a) Waterfall model - (b) Prototyping model - (c) Spiral model - (d) None of the above

Correct Answer: (a) Waterfall model

Technical Rationale: When requirements are completely understood, stable, and clearly defined upfront, the linear sequential waterfall model is best suited because it offers clean, disciplined phase execution.

ملاحظة الطالب والفخ الامتحاني: - مصفوفة متطلبات الشريحة 37: "Are requirements easily understandable and defined? → Yes :Waterfall ,Yes :RAD". الخيار الوحيد هنا بين النماذج المطروحة هو Waterfall.

Source: K.K. Aggarwal & Singh (2007) Slide 37; Slide 43 (Q2.11).

Question 2.12

If requirements are frequently changing, which model is to be selected? - (a) Waterfall model - (b) Prototyping model - (c) RAD model - (d) Iterative enhancement model

Correct Answer: (b) Prototyping model

Technical Rationale: When requirements change frequently or are unstable, the prototyping model is selected to construct rapid throw-away versions that allow users to evaluate features and stabilize requirements before full development.

> **ملاحظة الطالب والفخ الامتحاني:** - في مصفوفة الشريحة 37: "Do we change requirements quite often? → Prototype: Yes, Spiral: Yes". بما أن Spiral غير موجود في الخيارات، فالجواب هو Prototyping model.

Source: K.K. Aggarwal & Singh (2007) Slide 37; Slide 43 (Q2.12).

Question 2.13

If user participation is available, which model is to be chosen? - (a) Waterfall model - (b) Iterative enhancement model - (c) Spiral model - (d) RAD model

Correct Answer: (d) RAD model

Technical Rationale: The RAD model strictly mandates active and continuous user involvement across all phases (Joint Application Development) to evaluate progress and guide component construction.

> **ملاحظة الطالب والفخ الامتحاني:** - الشريحة 39 صريحة: "User involvement in all phases → Prototype: Yes, RAD: Yes". بين الخيارات المعروضة، RAD هو الأنسب للاشتراك المكثف.

Source: K.K. Aggarwal & Singh (2007) Slide 12, 39; Slide 43 (Q2.13).

Question 2.14

If limited user participation is available, which model is to be selected? - (a) Waterfall model - (b) Spiral model - (c) Iterative enhancement model - (d) Any of the above

Correct Answer: (d) Any of the above

Technical Rationale: According to the selection matrix based on user involvement, Waterfall, Spiral, and Iterative Enhancement all permit execution under limited user participation (unlike RAD, which strictly fails without heavy user involvement).

> **ملاحظة الطالب والفخ الامتحاني:** - الشريحة 39: سطر "Limited user participation" يضع علامة Yes أمام Waterfall و Iterative enhancement و Evolutionary و Spiral، ويضع No أمام RAD و Prototype. لذلك الخيار الصحيح هو "any of the above".

Source: K.K. Aggarwal & Singh (2007) Slide 39; Slide 43 (Q2.14).

Question 2.15

If project is the enhancement of existing system, which model is best suited? - (a) Waterfall model - (b) Prototyping model - (c) Iterative enhancement model - (d) Spiral model

Correct Answer: (c) Iterative enhancement model

Technical Rationale: When a project represents an upgrade or enhancement to an existing, already operational software system, the iterative enhancement model is designed specifically to deliver added capabilities across successive increments.

> **ملاحظة الطالب والفخ الامتحاني:** - الشريحة 40: "Iterative → Project is the enhancement of the existing system: Yes ,Evolutionary :Yes ,RAD :Yes". بين الخيارات المتاحة، Iterative enhancement هو الخيار المطابق.

Source: K.K. Aggarwal & Singh (2007) Slide 40; Slide 43 (Q2.15).

Question 2.16

Which one is the most important feature of spiral model? - (a) Quality management - (b) Risk management - (c) Performance management - (d) Efficiency management

Correct Answer: (b) Risk management

Technical Rationale: Risk management is the foundational, indispensable characteristic of Boehm's spiral model; the process is explicitly designed to identify and mitigate risks in every cycle before incurring further costs.

> **ملاحظة الطالب والفخ الامتحاني:** - النموذج الحلزوني يُعرف بأنه Risk-Driven Process، وإدارة المخاطر هي وظيفته الجوهرية.

Source: K.K. Aggarwal & Singh (2007) Slide 19, 21; Slide 44 (Q2.16).

Question 2.17

Most suitable model for new technology that is not well understood is: - (a) Waterfall model - (b) RAD model - (c) Iterative enhancement model - (d) Evolutionary development model

Correct Answer: (d) Evolutionary development model

Technical Rationale: The evolutionary development model is recommended when a project utilizes novel or poorly understood technology, as it enables developers to explore technical feasibility iteratively and evolve functionality by category.

> **ملاحظة الطالب والفخ الامتحاني:** - نص الشريحة 15: "This model is useful for projects using new technology that is not well understood." احذر اختيار RAD لأن RAD يفشل إذا كانت التكنولوجيا جديدة وغير مجربة.

Source: K.K. Aggarwal & Singh (2007) Slide 15; Slide 44 (Q2.17).

Question 2.18

Statistically, the maximum percentage of errors belong to the following phase of SDLC: - (a) Coding - (b) Design - (c) Specifications - (d) Installation and maintenance

Correct Answer: (c) Specifications

ملاحظة الطالب والفخ الامتحاني: - مفتاح الكتاب الرسمي (ص. 499) والشريحة 44 يحددان الخيار (c) كإجابة معتمدة للامتحان. - **حدود الدليل:** لا يورد متن الفصل أي جدول إحصائي أو مسح ميداني يثبت نسب الأخطاء بين المراحل؛ لذا يُحفظ المفتاح للامتحان دون اعتباره حقيقة إحصائية مثبتة في المتن.

Source: K.K. Aggarwal & Singh (2007) Slide 44 (Q2.18); Official Key p. 499.

Question 2.19

Which phase is not available in software life cycle? - (a) Coding - (b) Testing - (c) Maintenance - (d) Abstraction

Correct Answer: (d) Abstraction

Technical Rationale: Coding, Testing, and Maintenance are canonical operational phases of software life cycles. Abstraction is a fundamental software engineering principle and mental technique, not a designated chronological SDLC phase.

ملاحظة الطالب والفخ الامتحاني: - التجريد (Abstraction) مفهوم تصميمي وهندسي يُمارس في التحليل والتصميم، وليس مرحلة زمنية مستقلة في دورة الحياة.

Source: K.K. Aggarwal & Singh (2007) Slide 3; Slide 44 (Q2.19).

Question 2.20

The development is supposed to proceed linearly through the phases in: - (a) Spiral model - (b) Waterfall model - (c) Prototyping model - (d) None of the above

Correct Answer: (b) Waterfall model

Technical Rationale: The waterfall model is inherently structured as a linear-sequential process where development is assumed to flow in a strict, unbranching cascade from one phase to the next.

ملاحظة الطالب والفخ الامتحاني: - نموذج الشلال هو النموذج الخطي النموذجي (Linear Sequential Model). النماذج الأخرى (الحلزوني والبروتوتايب) دائرية أو تكرارية.

Source: B.B. Agarwal et al. (2010), §2.2, p. 36; K.K. Aggarwal & Singh (2007) Slide 6; Slide 44 (Q2.20).

Question 2.21

Unified process is maintained by: - (a) Infosys - (b) Rational software corporation - (c) SUN Microsystems - (d) None of the above

Correct Answer: (b) Rational software corporation

Technical Rationale: The Unified Process was enhanced, maintained, and productized by the Rational Software Corporation (which was subsequently acquired by IBM), and is commonly known as the Rational Unified Process (RUP).

ملاحظة الطالب والفخ الامتحاني: - نص السلايد 23: "Corporation Maintained and enhanced by Rational Software (RUP) Process and thus referred to as Rational Unified".

Source: K.K. Aggarwal & Singh (2007) Slide 23; Slide 45 (Q2.21).

Question 2.22

Unified process is: - (a) Iterative - (b) Incremental - (c) Evolutionary - (d) All of the above

Correct Answer: (d) All of the above

Technical Rationale: The Unified Process integrates multiple paradigms: it is iterative (structured as successive iterations), incremental (delivering expanded capability cycle by cycle), and evolutionary (advancing the architectural baseline over time).

ملاحظة الطالب والفخ الامتحاني: - يتميز الـProcess Unified بأنه يجمع الخصائص الثلاث معاً: Iterative و Incremental و Evolutionary، لذلك الجواب هو All of the above.

Source: K.K. Aggarwal & Singh (2007) Slide 23, 26; Slide 45 (Q2.22).

Question 2.23

Who is not in the team of Unified process development? - (a) I. Jacobson - (b) G. Booch - (c) B. Boehm - (d) J. Rumbaugh

Correct Answer: (c) B. Boehm

Technical Rationale: The Unified Process and UML were created by the "Three Amigos": Ivar Jacobson, Grady Booch, and James Rumbaugh. Barry Boehm is the creator of the Spiral Model and COCOMO, not an author of the Unified Process.

ملاحظة الطالب والفخ الامتحاني: - نص السلايد 23 يذكر الثلاثي: "Developed by I. Jacobson, G. Booch and J. Rumbaugh". باري بوم (Boehm) هو رائد النموذج الحلزوني وليس من مبتكري UP.

Source: K.K. Aggarwal & Singh (2007) Slide 23; Slide 45 (Q2.23).

Question 2.24

How many phases are in the unified process? - (a) 4 - (b) 5 - (c) 2 - (d) None of the above

Correct Answer: (a) 4

Technical Rationale: The Unified Process divides the life cycle into exactly four chronological phases: Inception, Elaboration, Construction, and Transition.

ملاحظة الطالب والفخ الامتحاني: - عدد مراحل UP هو 4 حصراً: Inception ← Elaboration ← Construction ← Transition.

Source: K.K. Aggarwal & Singh (2007) Slide 24; Slide 45 (Q2.24).

Question 2.25

The outcome of construction phase can be treated as: - (a) Product release - (b) Beta release - (c) Alpha release - (d) All of the above

Correct Answer: (b) Beta release

Technical Rationale: The milestone of the Construction phase is Initial Operational Capability (IOC), producing an operational software product release distributed for beta testing and user evaluation prior to final transition.

ملاحظة الطالب والفخ الامتحاني: - مرحلة Transition تبدأ باختبارات البيتا (Beta testing)، مما يعني أن ناتج مرحلة Construction هو الإصدار التجريبي المكتمل وظيفياً (Beta release). بينما الإصدار النهائي المكتمل للمستهلك (Product release) هو ناتج Transition.

Source: K.K. Aggarwal & Singh (2007) Slides 32–35; Slide 45 (Q2.25).

Retrieval set — Focused Active Recall Drill

1 What fundamental distinction separates an SDLC process model from an organizational framework like CMM?

An SDLC process model (such as Waterfall, Spiral, or RAD) prescribes the chronological phases, technical activities, and deliverables for engineering software. In contrast, the Capability Maturity Model (CMM) is an organizational appraisal framework that assesses the maturity and process management capability of the organization as a whole, rather than governing development phases.

Source: K.K. Aggarwal & Singh (2007) Slide 41 (Q2.3).

2 Why does the Waterfall model require requirements to be completely defined and frozen before design and coding commence?

Because development is structured to proceed strictly linearly without backward iteration. Once a phase is completed, subsequent stages depend entirely on its baseline deliverables, making late requirement changes disruptive and difficult to accommodate within a linear cascade.

Source: K.K. Aggarwal & Singh (2007) Slides 6, 8, 9; Slide 42 (Q2.8); Slide 44 (Q2.20).